جامعة مصر للمعلوماتية
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OF INFORMATICS

Computing and Information Sciences
Data Analysis Course

Study Time (Day VS Night)

& Academic Performance Among University Students

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Introduction

Academic performance is influenced by many factors, including study habits, sleep patterns, and time management. One of the most debated topics among students is whether studying during the day or at night leads to better academic results. Some students believe they are more productive at night due to fewer distractions, while others prefer studying during the day when they feel more focused and energetic.

This study aims to explore the relationship between study time (day vs. night) and academic performance among university students. Understanding this relationship can help students improve their study habits and achieve better academic outcomes.

Research Question

Does studying at night or during the day affect academic performance among university students?

Hypothesis

Students who study during the day perform better academically than those who study at night.

Methodology

1. Population of Interest:

University students from different universities, years, and majors.

2. Sampling Method:

A convenience sampling method was used to collect data for this study. The survey was distributed by all members of the project team, who shared the form with their friends and relatives currently enrolled in university. This approach allowed for quick and efficient data collection. Furthermore, responses were collected from students across different universities, academic years, and majors, which helped improve the diversity of the sample and make the results more representative of the target population.

3. Bias Identification:

- In designing this survey, several potential sources of bias were identified and addressed. One possible source is **sampling bias**, as the survey was distributed using convenience sampling through the project team members, who shared it with their friends and relatives currently enrolled in university. This may lead to overrepresentation of certain social groups. To reduce this bias, the survey was shared by multiple team members and included students from different universities, academic years, and majors, helping to improve the diversity of the sample.

- Another potential source of bias is **self-reporting bias**, since participants provided their own information regarding GPA, study hours, and sleep habits. These responses may not always be fully accurate due to overestimation or underestimation. However, the survey was conducted anonymously, which helps reduce this bias by encouraging participants to answer more honestly without fear of judgment.

- **Response bias** may also occur if some participants did not answer the questions carefully. To minimize this, the survey questions were designed to be simple, clear, and neutral, avoiding any leading or suggestive wording. Overall, while some bias may still exist, efforts were made to improve the representativeness and reliability of the collected data.

Survey Questions:

1. When do you usually study the most?
2. On average, how many hours do you study per day?
3. What is your current GPA?
4. What is your current year of study?
5. How many hours do you sleep on average per night?
6. Do you believe your study time (day/night) affects your academic performance?

Online survey link:
https://docs.google.com/forms/d/e/1FAIpQLSdnR160X_i2nCFzdRER6LE6Z1Wb957I_sxIN-HOY79X6IbjbA/viewform?usp=publish-editor

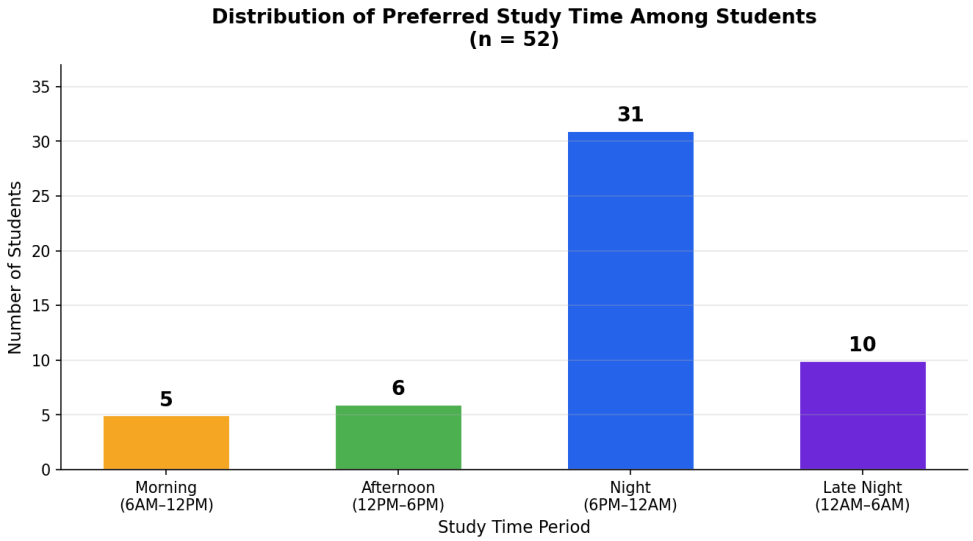
Number of samples collected: 52

Analysis

To analyse the data collected from this survey, we calculated basic descriptive statistics such as the mean, median, and mode. We also created visual representations of the data using charts and graphs to help identify trends and patterns.

Variable	Mean	Median	Mode
GPA	2.80	3.25	3.25
Study Hours/Day	3.0 hrs	3.5 hrs	3.5 hrs
Sleep Hours/Night	6.63 hrs	6.5 hrs	6.5 hrs

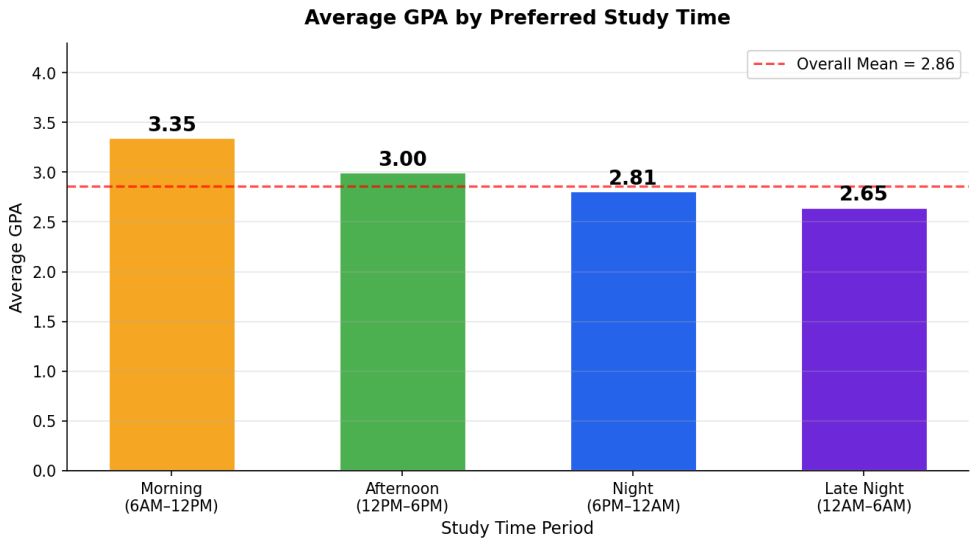
Study Time Distribution:



Most students prefer studying at night, particularly between 6 PM and 12 AM, making it the most common study time among participants.

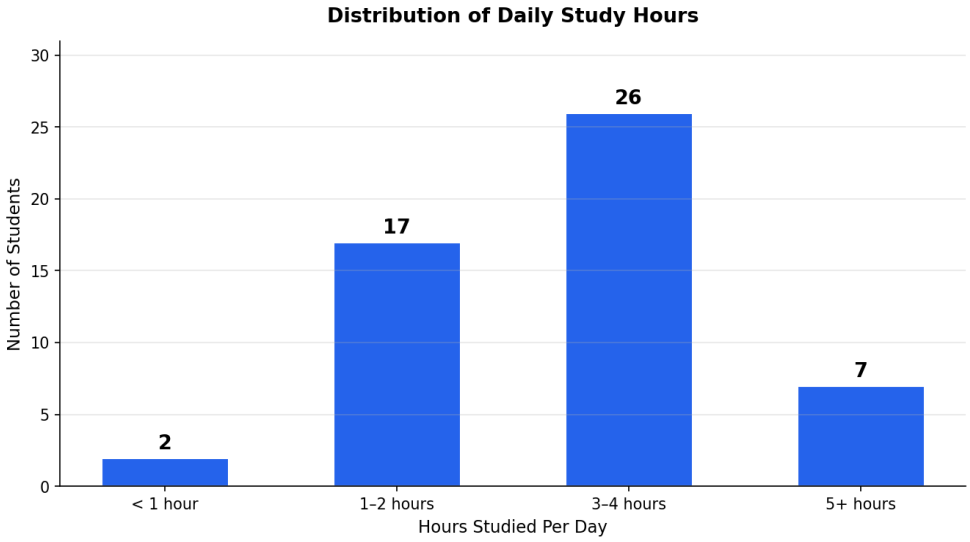
Night (6PM–12AM): 31 students, Late Night (12AM–6AM): 10 students, Afternoon: 6 students, Morning: 5 students.

Average GPA by Study Time:



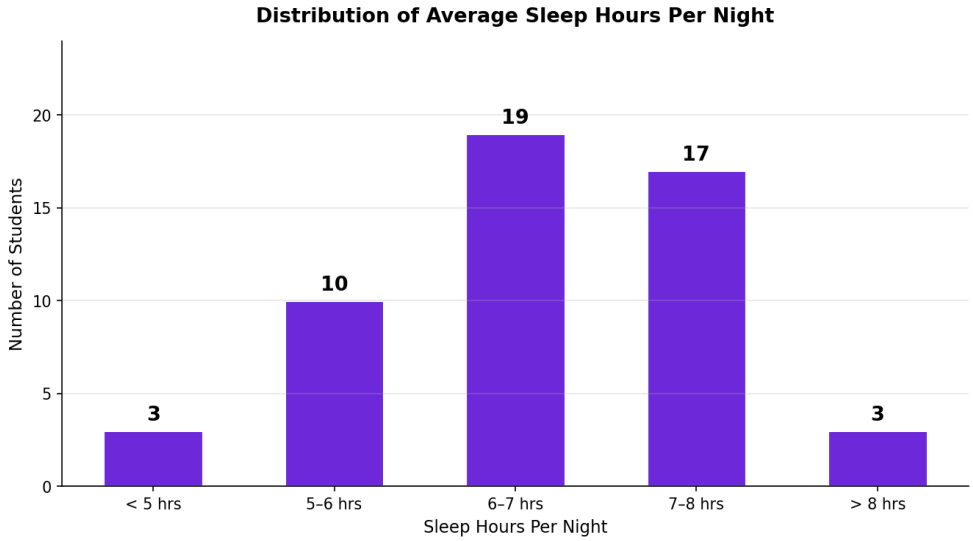
This chart compares the average GPA of students based on their preferred study time. It is clearly observed that students who study in the morning have the highest average GPA (3.35), while those who study late at night have the lowest average GPA (2.65). This indicates a noticeable decline in academic performance as study time shifts later into the night. These findings strongly support the hypothesis that studying earlier in the day leads to better academic performance.

Study Hours Distribution:



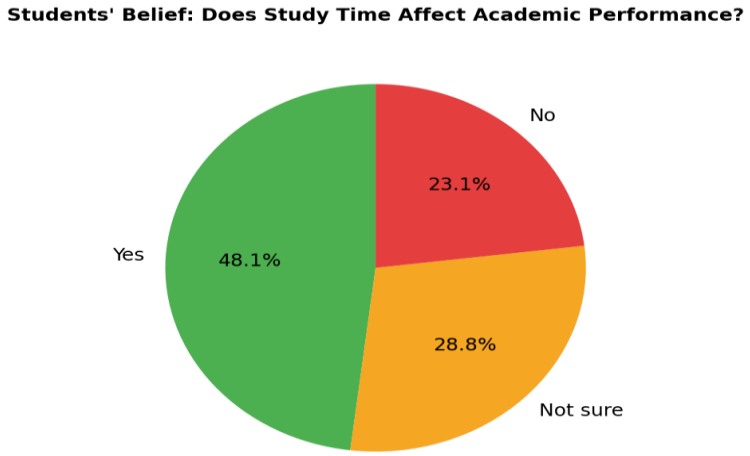
The analysis shows a slight positive relationship between study hours and GPA. Students who study more tend to achieve better academic results; however, the relationship is not very strong, suggesting that study quality and timing may be more important than quantity alone.

Sleep Hours Distribution:



Sleep duration appears to play an important role in academic performance. Students who sleep around 6 to 8 hours tend to have better GPAs compared to those with very low sleep hours, indicating that proper rest contributes to better focus and learning efficiency.

Students' Opinion on Study Time:



Most students believe that their study time (day or night) affects their academic performance. This aligns with the actual data findings, which show differences in GPA based on study timing.

Conclusion

Based on the analysis, there is a clear relationship between study time and academic performance. Students who study earlier in the day, particularly in the morning, achieve higher GPAs compared to those who study late at night.

The data shows a gradual decrease in GPA as study time shifts from morning to late night. Therefore, the hypothesis is supported by the findings. However, other factors such as sleep duration and study hours also contribute to academic success.

Critical Reflection

Several issues and limitations were identified during the data collection and analysis process.

First, because the survey was distributed through personal networks (convenience sampling), the sample may over-represent students from similar social backgrounds and may not fully represent all university students.

Second, the sample sizes for daytime study groups were very small (only 5 for Morning and 6 for Afternoon), meaning their average GPAs may not be statistically robust.

Third, participants self-reported their GPA, study hours, and sleep habits from memory, which introduces the possibility of inaccuracy.

Finally, because GPA and study/sleep hours were collected as categorical ranges (e.g., "3.0–3.5") rather than exact values, we had to use midpoint approximations for our numerical calculations, which introduces a degree of imprecision.

In future research, requiring respondents to input exact numerical values would allow for more accurate data analysis.